

Aditya Chander

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EDUCATION

- 2019-2025 *Yale University*
PhD candidate, Music Theory (Advisors: Ian Quinn, Richard Aslin, Daniel Walden)
Dissertation: “Reciprocal relationships between musical expectation and repetition structure”
Qualifying exams passed with Distinction, August 2021
Distinguished Dissertation, April 2025
- 2017-2019 *Stanford University*
MA, Music, Science and Technology (Advisor: Takako Fujioka)
Capstone project: “Violinists employ more expressive gesture and timing around global musical resolutions: A motion capture study” (published in *Music Perception*)
- 2014-2017 *University of Cambridge*
BA (Hons), Music
Double First-Class Honors with Distinction

PROFESSIONAL EXPERIENCE

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|-------------------------|---|
| July 2025-present | Research Scientist, Upstart |
| June 2024-present | Project Mentor, Erdos Institute for Data Science |
| September 2023-present | Violinist, Boston Philharmonic Orchestra |
| June-August 2023 | Data Scientist, Eventellect |
| September 2019-May 2023 | Concertmaster/principal violinist, Yale Symphony Orchestra |
| March 2021-present | Writer, BBC Orchestras and Choirs |
| June 2020-July 2022 | Instructor and Curriculum Developer, Inspirit AI |
| October 2014-June 2017 | Concertmaster/principal violinist and Publicity Officer, Cambridge University Chamber Orchestra |

HONORS / AWARDS

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| 2024 | Gorilla Research Grant (https://gorilla.sc), Honorable Mention (received twice; equivalent of \$300) |
| 2024 | Subvention Grant, Grant Hagan Society (Yale University, \$380) |
| 2023 | Cultural Grant, Wesleyan/Yale/Yogyakarta Palace Faculty Exchange (American-Indonesian Cultural and Education Foundation, \$2000) |

- 2023 Best Essay in Music Informatics Scholarship for “Expectation adaptation for rare cadences in music: Item order matters in repetition priming” (Chander & Aslin, 2023, *Cognition*), awarded by the Music Informatics Interest Group, Society for Music Theory
- 2023 Travel Award, International Conference on Music Perception and Cognition (ICMPC)
- 2023 Shortlisted for Young Researcher Award, ICMPC
- 2022-2024 Travel Awards from Graduate Student Assembly, Yale University (x3, \$800 each)
- 2022 Best Data Science Project (Podcast Recommender App), Erdős Institute for Data Science
- 2022 Grant from Allen Forte Research Fund (Yale University, \$3000)
- 2020 Honorable Mention, World Bach Competition (Boulder Bach Festival; Boulder, CO; violin)
- 2019-2021 Richard J. Franke Fellowship (Yale University)
- 2017-2019 Denning Family Fellowship; Christensen Fellowship; & Friends of Music Scholarship (Stanford University)
- 2014-2017 College and Bundy Scholarship; Lincoln Award; College, Lincoln and Dorothy Kolbert Prizes for Music; & Mynors Bright Prize (Magdalene College, University of Cambridge)
- 2014 Finalist, National Center for Early Music Young Composers’ Competition (York, United Kingdom)
- 2013 Honorable Mention, The Times Christmas Carol Composition Competition (United Kingdom)
- 2012 Highly Commended, BBC Inspire Young Composers’ Competition (London, United Kingdom)

MANUSCRIPTS UNDER REVIEW & IN PROGRESS

Nerness, Barbara, Noah Fram, Kunwoo Kim, **Aditya Chander**, Cara Turnbull, Elena Georgieva, Sebastian James, Matt Wright, and Takako Fujioka. “Alpha EEG oscillations during turn-taking piano duet performance reflect improvisational creativity, musical role, and partner task similarity.” Under review, *Cerebral Cortex*.

Chander, Aditya. “Comparing perceptual theories of large-scale musical repetition structure: a corpus study.” Article in prep.

Quinn, Ian, **Aditya Chander**, and Stefanie Acevedo. “Tonicity and information in popular music harmony.” Invited chapter in prep for *The Oxford Handbook of Music and Corpus Studies*, edited by Daniel Shanahan, J. Ashley Burgoyne, and Ian Quinn. Oxford University Press.

PEER-REVIEWED JOURNAL ARTICLES

- 2025. Jimenez, Ivan, Tuire Kuusi, Juha Ojala, Isabella Czedik-Eysenberg, Peter Harrison, and **Aditya Chander**. “Effect of timbre on goodness-of-fit ratings of chord successions in non-cadential contexts.” *Music Perception*. <https://doi.org/10.1525/mp.2025.2397481>
- 2024. Kim, Kunwoo, Noah Fram, Barbara Nerness, Cara Turnbull, **Aditya Chander**, Elena Georgieva, Sebastian James, Matt Wright, and Takako Fujioka. “Performance monitoring of improvisation and score-

playing in a turn-taking piano duet: An EEG study using altered auditory feedback.” *Psychophysiology*. <https://doi.org/10.1111/psyp.14704>.

2023. **Chander, Aditya**, and Richard Aslin. ‘Expectation adaptation for rare cadences in music: Item order matters in repetition priming’. *Cognition* 240 (November): 13pp. <https://doi.org/10.1016/j.cognition.2023.105601>.
2022. **Chander, Aditya**, Madeline Huberth, Stacey Davis, Samantha Silverstein, and Takako Fujioka. ‘Violinists employ more expressive gesture and timing around global musical resolutions: A motion capture study’. *Music Perception* 39 (3): 268–88. <https://doi.org/10.1525/mp.2022.39.3.268>.

PUBLISHED CONFERENCE PROCEEDINGS

2023. **Chander, Aditya**, and Ian Quinn. 2023. “The decline of harmonic schemata in popular music chord loops.” In *Proceedings of the 17th International Conference on Music Perception and Cognition*, 34-39. Tokyo, Japan.
2023. Acevedo, Stefanie, **Aditya Chander**, and Ian Quinn. “Tonicity and information in three pop harmony corpora.” In *Proceedings of the 17th International Conference on Music Perception and Cognition*, 42-47. Tokyo, Japan.

CONFERENCE PRESENTATIONS AND POSTERS

2025. **Chander, Aditya**, and Richard Aslin. “Statistical learning for conventional vs. characteristic patterns.” International Conference on Music Perception and Cognition, São Paulo, Brazil.
2024. Chander, Aditya. “Learning and memory for conventional vs. characteristic patterns.” Music Cognition Interest Group, Annual Meeting of the Society for Music Theory, Jacksonville, FL.
2024. Chander, Aditya. “Perceptual influences on large-scale musical repetition structure: a corpus study.” Society for Music Perception and Cognition Conference, Banff, AB, Canada.
2024. **Chander, Aditya**, and Richard Aslin. “Modelling musical coherence in the repetition structure of novel compositions.” Society for Music Perception and Cognition Conference, Banff, AB, Canada.
2024. Sailor, Malcolm, and **Aditya Chander**. “Enharmonically equivalent spelling distinctions influence perceived consonance.” Society for Music Perception and Cognition Conference, Banff, AB, Canada.
2024. Arthur, Claire, **Aditya Chander**, and Farshad Jafari. “Modelling continuous emotion ratings with surface-level, structural, and information-theoretic predictors.” Society for Music Perception and Cognition Conference, Banff, AB, Canada.
2024. Jimenez, Ivan, Tuire Kuusi, Juha Ojala, Isabella Czedik-Eysenberg, Peter Harrison, and **Aditya Chander**. “Goodness-of-fit ratings of chords in non-cadential contexts can be affected by timbre.” Twelfth Triennial Conference of the European Society for the Cognitive Sciences of Music, York, UK.
2024. Chander, Aditya. “Perceptual perspectives on large-scale musical repetition structure: a corpus study.” New England Conference of Music Theorists, Boston, MA.

2023. **Chander, Aditya**, and Ian Quinn. "The decline of harmonic schemata in popular music chord loops." International Conference on Music Perception and Cognition, Tokyo, Japan.
2023. Acevedo, Stefanie, **Aditya Chander**, and Ian Quinn. "Tonicity and information in three pop harmony corpora." International Conference on Music Perception and Cognition, Tokyo, Japan.
2023. Fram, Noah, **Aditya Chander**, Barbara Nerness, Kunwoo Kim, Cara Turnbull, Elena Georgieva, Sebastian James, Matt Wright and Takako Fujioka "Neural alpha oscillations reveal interplay between creative ideation and cooperation in joint improvisation during piano-duet turn-taking." International Conference on Music Perception and Cognition, Tokyo, Japan.
2023. Kim, Kunwoo, Noah Fram, Barbara Nerness, Cara Turnbull, **Aditya Chander**, Elena Georgieva, Sebastian James, Matt Wright and Takako Fujioka. "Performance monitoring through altered auditory feedback in musical actions with score and improvisation: A turn-taking piano-duet EEG study." International Conference on Music Perception and Cognition, Tokyo, Japan.
2023. Glotzer, Giacomo, **Aditya Chander**, and Robb Rutledge. "Statistical learning informs predictive models of musical pleasure." International Conference on Music Perception and Cognition, Tokyo, Japan.
2023. Glotzer, Giacomo, **Aditya Chander**, and Robb Rutledge. "Variable-order Markov chains inform predictive models of musical value." Computational Psychiatry Conference, Dublin, Ireland.
2023. Chander, Aditya. "Harmonic expectation and distinctive chord loops in two popular music corpora." Music Theory Midwest Annual Meeting, Winnipeg, MB, Canada.
2022. **Chander, Aditya**, and Richard Aslin. "Repetition priming and dynamic expectation of cadences." Society for Music Perception and Cognition Conference, Portland, OR.
2022. Chander, Aditya. "Repetition priming and dynamic expectation of cadences." University of Connecticut Graduate Music Conference, Storrs, CT.
2022. Fram, Noah, **Aditya Chander**, Kunwoo Kim, Barbara Nerness, Cara Turnbull, Elena Georgieva, Sebastian James, Matthew Wright and Takako Fujioka. "The exchange of musical ideas in duet turn-taking improvisation is related to empathy and its between-partner difference." Society for Music Perception and Cognition Conference, Portland, OR.
2022. **Chander, Aditya**, and Richard Aslin. "Rapid expectation adaptation for rare cadences in music." International Conference on Interdisciplinary Advances in Statistical Learning, Donostia-San Sebastián, Basque Country, Spain.
2021. Chander, Aditya. "Analyzing hemiolas using the Discrete Fourier Transform." SysMus21 (International Conference of Students of Systematic Musicology), Aarhus, Denmark.
2021. Chander, Aditya. "Analyzing hemiolas using the Discrete Fourier Transform." Society for Music Theory Annual Meeting, online conference.
2020. Nerness, Barbara, Noah Fram, Kunwoo Kim, **Aditya Chander**, Cara Turnbull, Elena Georgieva, Sebastian James, Matthew Wright, and Takako Fujioka. "Neural alpha oscillations during turn-taking piano duet index creative thinking and engagement to the partner's action." Auditory Perception, Cognition, and Action Meeting, online conference.
2020. Nerness, Barbara, Noah Fram, Kunwoo Kim, **Aditya Chander**, Cara Turnbull, Elena Georgieva, Sebastian James, Matthew Wright, and Takako Fujioka. "Neural alpha oscillations during turn-taking

piano duet index creative thinking and engagement to the partner's action." Cognitive Neuroscience Society, online conference.

2019. **Chander, Aditya**, Madeline Huberth, Stacey Davis, Samantha Silverstein, and Takako Fujioka. "Violinists employ more expressive gesture around musical resolutions: A motion capture study." Society for Music Perception and Cognition Conference, New York, NY.

INVITED TALKS

April 2025. "Perceptual theories of repetition in musical form." Hartt School, University of Hartford, Hartford, CT.

March 2025. "Perceptual theories of repetition in musical form." MIND Lab, Northeastern University, Boston, MA.

October 2024. "Learning and memory for conventional vs. characteristic patterns." MIND Lab, Northeastern University, Boston, MA.

June 2024. "Modelling structures of thematic repetition in music." Music Cognition Lab, Queen Mary University of London.

February 2024. "How does perception influence large-scale musical repetition?" Works in Progress, Department of Music, Yale University.

October 2023. "Dynamic expectation adaptation in harmony." MIND Lab, Northeastern University, Boston, MA.

September 2022. "Rapid expectation adaptation for rare cadences." Music Cognition Lab, Queen Mary University of London.

TEACHING EXPERIENCE

Instructor, MUSI 211 (Systematic Music Theory: 1800-present), Yale University, Spring 2025.

Instructor, MUSI 210 (Counterpoint, Harmony and Form: 1500-1800), Yale University, Spring 2023.

Teaching Fellow, MUSI 110 (Elements of Musical Pitch and Time), Yale University, Fall 2022.

Teaching Fellow, MUSI 131 (Introduction to the History of Western Music: 1800 to the Present), Yale University, Spring 2022.

Teaching Fellow, MUSI 216 (Meter, Rhythm, Musical Time), Yale University, Fall 2021.

Teaching Fellow, CPSC 477/577 (Natural Language Processing), Yale University, Spring 2021.

Instructor and Curriculum Developer (Introduction to Python & Music Recommenders project), Inspirit AI, May 2020-July 2022. (*Inspirit AI is a high-school intensive camp in machine learning and artificial intelligence.*)

Senior Section Leader, CS106A/AP/B/X (Programming Methodology and Abstractions), Stanford University. Taught in the following quarters: Fall 2018, Winter 2018, Spring 2019, Summer 2019.

MENTORSHIP EXPERIENCE

Students mentored:

- **Nicholas Boettcher** (Yale BS, Cognitive Science, expected 2025), research assistant in LLAMB lab.
- **Giacomo Glotzer** (Yale BS, Neuroscience, 2023), co-advised senior thesis with Prof. Robb Rutledge. Now: PhD student, Neuroscience, Rockefeller University.

JOURNAL REVIEWING

Ad-hoc for *Journal of Music Theory*; *Transactions of the International Society for Music Information Retrieval*; *Language Learning*; *Cognition*; *PLOS ONE*; *Scientific Reports*.

SERVICE WORK

Co-Chair, Music Cognition Interest Group, Society for Music Theory (2024-Present). Elected at Annual Meeting of SMT in November 2024.

Grant Hagan Society, Department of Music, Yale University (<https://campuspress.yale.edu/granthagan/>): **Member at Large (2023-present), Co-chair (2021-2023), Webmaster (2019-2021).**

Events organized: Navigating the (Music) Academic Job Market as a Person of Color (2020) – communications officer; Underrepresented in the Archives (2020) – event scribe; Critical Perspectives on Diversity Committees (2021) – moderator; Anti-Racist Pedagogy Inside and Outside the University Music Classroom (2023)* – head coordinator.

*Successfully applied for **\$3700 of funding** for this event from several organizations at Yale: Graduate and Professional Student Senate; Center for Race, Indigeneity and Transnational Migration; Graduate School of Arts and Sciences Dean's Fund; Belonging @ Yale; Department of Music.

Teaching Fellow Selection Committee, Department of Music, Yale University: Student Representative (provided input on appropriate selection of teaching fellows in the Department of Music for the 2023-2024 academic year)

Grad-Fac committee, Department of Music, Yale University: First and Second-Year Representative (2020-2021). Grad-Fac is the information bridge between the faculty and the graduate students in the Department of Music at Yale University.

Regular responsibilities: Meeting with the chair of the department once a month (more frequently during the height of the pandemic); I spearheaded weekly Zoom coffee hours during the pandemic and regular Town Halls each semester.

Other actions: Revised the language of the Graduate Student Handbook to be more inclusive; co-convened the Anti-Racist Reading Group in the Department of Music.

MEMBERSHIPS / AFFILIATIONS

Society for Music Perception and Cognition, Society for Music Theory, Music Theory Midwest

TECHNICAL SKILLS

Notation software: Sibelius, MuseScore

Audio editing software: Audacity

Programming/markup languages:

- Most commonly used: Python
 - Machine learning frameworks: scikit-learn, keras
 - Other packages: numpy, pandas, matplotlib
- Semi-regularly used: R, LaTeX, JavaScript, MATLAB, Common Lisp (IDyOM)
- Previous experience: Java, C++
- Other tools: git, bash
- Elementary proficiency: ChuckK, MaxMSP, HTML/CSS

References

Ian Quinn

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Richard Aslin

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